Introduction to Analysis I

Lecture notes, Chapters 1–5 and Appendix

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Chapter 1. Real Numbers and Limits of Sequences

Analysis rests on a precise description of the real numbers. We take $\mathbb{R}$ as given by the field axioms, the order axioms and the completeness axiom, and derive everything else from them; Appendix A constructs a field satisfying the axioms. Book numbers are given in brackets.

1.1 Field and order axioms

Definition (Ordered field)

A **field** is a set F with operations $+$ and $\cdot$, both associative and commutative, with $\cdot$ distributive over $+$, identities $0 \neq 1$, additive inverses $-a$, and multiplicative inverses a^{-1} for $a \neq 0$ (axioms (F1)–(F9)). A field is **ordered** if there is a subset P (the positive elements) with (O1) $a, b \in P \Rightarrow a+b \in P$; (O2) $a, b \in P \Rightarrow ab \in P$; (O3) for each a exactly one of $a \in P$, $a = 0$, $-a \in P$ holds. We write $a < b$ (or $b > a$) if $b - a \in P$, and $a \leq b$ if $a < b$ or $a = b$.

Proposition 1.1 (1.1.1)

In a field:
(a) $-(-a) = a$, and $(a^{-1})^{-1} = a$ for $a \neq 0$;
(b) $a + b = a + c \Rightarrow b = c$, and $ab = ac$, $a \neq 0 \Rightarrow b = c$;
(c) $ab = 0 \iff a = 0$ or $b = 0$;
(d) $a \cdot 0 = 0$, $(-a)b = -(ab)$, $(-a)(-b) = ab$;
(e) the identities and inverses are unique.

Proof. (e) If $0, 0'$ are additive identities, $0 = 0 + 0' = 0'$; if b, b' are inverses of a, $b = b + (a + b') = (b + a) + b' = b'$; likewise for $\cdot$.

(b) Add $-a$ to both sides (multiply by a^{-1}).

(a) a and $-(-a)$ are both additive inverses of $-a$; use (e).

(d) $a \cdot 0 + a \cdot 0 = a(0 + 0) = a \cdot 0 = a \cdot 0 + 0$, so $a \cdot 0 = 0$ by (b); $ab + (-a)b = (a - a)b = 0$ gives $(-a)b = -(ab)$, and then $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.

(c) If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ∎

Proposition 1.2 (1.1.2)

In an ordered field:
(a) $a \leq b$ and $b \leq a$ imply $a = b$;
(b) $a < b$, $b < c$ imply $a < c$;
(c) $a + b < a + c \iff b < c$;
(d) if $a > 0$ then $b < c \iff ab < ac$, and if $a < 0$ then $b < c \iff ab > ac$;
(e) $a^2 > 0$ for $a \neq 0$, in particular $1 > 0$;
(f) $a > 0 \Rightarrow a^{-1} > 0$, and $0 < a < b \Rightarrow 0 < b^{-1} < a^{-1}$;
(g) $a < b$, $c < d \Rightarrow a + c < b + d$;
(h) for $a, b \geq 0$: $a < b \iff a^2 < b^2$, and $a^2 = b^2 \Rightarrow a = b$.

Proof. (a) If $a \neq b$, both $b - a$ and $a - b = -(b - a)$ would be in P, against (O3).

(b) $c - a = (c - b) + (b - a) \in P$.

(c) $(a + c) - (a + b) = c - b$.

(d) $ac - ab = a(c-b)$; for $a > 0$ use (O2) and trichotomy, for $a < 0$ note $-a > 0$.

(e) If $a > 0$, $a^2 \in P$; if $a < 0$, $a^2 = (-a)^2 \in P$; and $1 = 1^2$.

(f) If $a^{-1} \leq 0$ then $1 = aa^{-1} \leq 0$; multiply $a < b$ by $a^{-1}b^{-1} > 0$.

(g) $(b+d) - (a+c) = (b-a) + (d-c)$.

(h) $b^2 - a^2 = (b-a)(b+a)$ and $b + a > 0$ when $a, b \geq 0$ are not both 0. ■

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. $\mathbb{C}$ is not: in an ordered field $-1 < 0 < 1 = 1^2$, but $i^2 = -1$ would have to be positive. No finite field is ordered, since $1, 1+1, 1+1+1, \ldots$ are strictly increasing.

The **absolute value** is $|a| = a$ for $a \geq 0$ and $|a| = -a$ for $a < 0$.

Proposition 1.3 (1.1.3)

(a) $|a| \geq 0$, and $|a| = 0 \iff a = 0$;
(b) $|ab| = |a||b|$;
(c) for $b \geq 0$, $|a| \leq b \iff -b \leq a \leq b$;
(d) $|a + b| \leq |a| + |b|$ (triangle inequality);
(e) $||a| - |b|| \leq |a - b|$.

Proof. (a), (b) are case checks.

(c) If $a \geq 0$, $|a| \leq b \iff a \leq b$, and $-b \leq 0 \leq a$ automatically; similarly for $a < 0$.

(d) From $-|a| \leq a \leq |a|$ and the same for b, $-(|a| + |b|) \leq a + b \leq |a| + |b|$; apply (c).

(e) $|a| = |(a-b) + b| \leq |a - b| + |b|$, and symmetrically. ■

1.2 The completeness axiom

$S \subset \mathbb{R}$ is **bounded above** if it has an **upper bound** u ($s \leq u$ for all $s \in S$). A **least upper bound** (**supremum**) $\sup S$ is an upper bound smaller than every other upper bound; it is unique if it exists. Lower bounds and $\inf S$ are defined dually.

Definition (Completeness axiom)

(C) Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Proposition 1.4 (1.2.1)

In an ordered field the following are equivalent:
(a) every nonempty set bounded above has a supremum;
(b) every nonempty set bounded below has an infimum.

Proof. (a)⇒(b): let $S \neq \emptyset$ be bounded below and L the set of lower bounds of S; $L \neq \emptyset$, and every $s \in S$ is an upper bound of L. Let $a = \sup L$. Each $s \in S$ is an upper bound of L, so $a \leq s$: $a \in L$. If $b \in L$ then $b \leq a$. So $a = \inf S$.

(b)⇒(a) is symmetric, or apply (b) to $-S = \{-s\}$. ■

A useful reformulation: $u = \sup S$ iff u is an upper bound and for every $\epsilon > 0$ there is $s \in S$ with $s > u - \epsilon$.

Proposition 1.5 (1.2.2)

$\mathbb{N}$ is not bounded above.

Proof. Otherwise $a = \sup \mathbb{N}$ exists; $a - 1$ is not an upper bound, so $a - 1 < n$ for some $n \in \mathbb{N}$, and then $n + 1 > a$ with $n + 1 \in \mathbb{N}$, a contradiction. ■

Corollary 1.6 (1.2.3)

$\inf\{1/n : n \in \mathbb{N}\} = 0$; that is, for every $\epsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \epsilon$.

Proof. 0 is a lower bound. If $\epsilon > 0$ were a lower bound, then $n \leq 1/\epsilon$ for all n, contradicting Proposition 1.5. ■

Corollary 1.7 (1.2.4, Archimedean property)

For $a > 0$ and $b \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $na > b$.

Proof. By Proposition 1.5 choose $n > b/a$. ■

Theorem 1.8 (1.2.5, density of the rationals)

If $a < b$ there is $r \in \mathbb{Q}$ with $a < r < b$.

Proof. First let $0 \leq a < b$. Choose n with $1/n < b - a$ (Corollary 1.6). The set $\{m \in \mathbb{N} : m/n > a\}$ is nonempty (Corollary 1.7); let m be its least element. Then $(m-1)/n \leq a$ (if $m = 1$ this holds as $a \geq 0$), so $a < m/n \leq a + 1/n < b$. If $a < 0$, choose $k \in \mathbb{N}$ with $k > -a$, find r between $a + k \geq 0$ and $b + k$, and take $r - k$. ■

The irrationals are also dense: apply the theorem to $a/\sqrt{2} < b/\sqrt{2}$ (using Proposition 1.9).

Proposition 1.9 (1.2.6)

For each $a > 0$ there is a unique $b > 0$ with $b^2 = a$.

Proof. Uniqueness: Proposition 1.2(h).

Existence: $S = \{x > 0 : x^2 < a\}$ is nonempty (it contains $t = a/(1 + a)$, since $t < 1$ and $t < a$ give $t^2 < t < a$) and bounded above by $1 + a$ (if $x > 1 + a$ then $x^2 > x > a$). Let $b = \sup S > 0$. If $b^2 < a$, for $0 < h < 1$ we have $(b+h)^2 \leq b^2 + h(2b+1) < a$ once $h < (a - b^2)/(2b+1)$, so $b + h \in S$, a contradiction. If $b^2 > a$, for $0 < h < b$, $(b - h)^2 > b^2 - 2bh > a$ once $h < (b^2 - a)/2b$; then $b - h$ is an upper bound of S (if $x \geq b - h$ then $x^2 \geq (b - h)^2 > a$), a contradiction. So $b^2 = a$. ■

In particular $\mathbb{Q}$ does not satisfy (C): $\{x \in \mathbb{Q} : x > 0, x^2 < 2\}$ has no rational supremum, as $\sqrt{2}$ is irrational.

Theorem 1.10 (1.2.7, uniqueness of the real field)

Let F and G be ordered fields satisfying the completeness axiom. There is a bijection $f : F \to G$ with $f(a + b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$, and $a < b \iff f(a) < f(b)$.

Proof. Each ordered field contains a copy of $\mathbb{Q}$: $n \mapsto 1 + \cdots + 1$, extended to $\mathbb{Z}$ and to quotients; this map $\mathbb{Q} \to F$ preserves operations and order (Proposition 1.2). Let $\phi : \mathbb{Q}_F \to \mathbb{Q}_G$ be the resulting isomorphism of the rational subfields. Both F and G are Archimedean (Corollary 1.7), so the rationals are dense in each (Theorem 1.8). Define

$$f(x) = \sup\{\phi(r) : r \in \mathbb{Q}_F,\ r < x\}.$$

The set is nonempty and bounded above (by $\phi(s)$ for any rational $s > x$), so $f(x)$ exists. Order: if $x < y$, pick rationals $x < r < s < y$; then $f(x) \leq \phi(r) < \phi(s) \leq f(y)$. Also $f(r) = \phi(r)$ for rational r (density). So f is strictly increasing, hence injective. Surjective: for $z \in G$ let $x = \sup\{r \in \mathbb{Q}_F : \phi(r) < z\}$; density in G and strict monotonicity show $f(x) = z$. Additivity:

for rationals $r < x$, $s < y$ we have $r + s < x + y$, so $\phi(r) + \phi(s) \le f(x + y)$, and taking suprema $f(x) + f(y) \le f(x + y)$; applying this to $-x, -y$ (note $f(-x) = -f(x)$, from the definition and density) gives equality. Multiplicativity for $x, y > 0$: products rs of positive rationals $r < x$, $s < y$ are exactly cofinal among rationals below xy, giving $f(xy) = f(x)f(y)$; extend to all signs using $f(-x) = -f(x)$. ■

So the axioms determine $\mathbb{R}$ up to isomorphism.

Proposition 1.11 (1.2.8)

For nonempty bounded $S, T \subset \mathbb{R}$, $\sup(S+T) = \sup S + \sup T$ and $\inf(S+T) = \inf S + \inf T$, where $S + T = \{s + t\}$. If S, T consist of positive numbers, $\sup(ST) = \sup S \cdot \sup T$ and $\inf(ST) = \inf S \cdot \inf T$.

Proof. $\sup S + \sup T$ is an upper bound of $S+T$. Given $\epsilon > 0$ choose $s > \sup S - \epsilon/2$, $t > \sup T - \epsilon/2$; then $s + t > \sup S + \sup T - \epsilon$. For products, $\sup S \sup T$ is an upper bound of ST; with $a = \sup S$, $b = \sup T$ and $0 < \delta < \min(a, b)$, choose $s > a - \delta$, $t > b - \delta$, so $st > (a-\delta)(b-\delta) = ab - \delta(a+b-\delta)$, which is arbitrarily close to ab. For infima, $\inf S \inf T$ is a lower bound of ST, and with $s < \inf S + \delta$, $t < \inf T + \delta$ we get $st < \inf S \inf T + \delta(\inf S + \inf T + \delta)$. ■

1.3 Limits of sequences

(x_n) **converges** to x, $\lim x_n = x$, if for every $\epsilon > 0$ there is N with $|x_n - x| < \epsilon$ for all $n \ge N$.

Proposition 1.12 (1.3.1)

Limits are unique.

Proof. If $x_n \to x$, $x_n \to y$ and $x \ne y$, take $\epsilon = |x - y|/2$ and $n \ge \max(N_1, N_2)$: $|x - y| \le |x - x_n| + |x_n - y| < 2\epsilon = |x - y|$. ■

Proposition 1.13 (1.3.2)

Every convergent sequence is bounded.

Proof. If $x_n \to x$, choose N with $|x_n - x| < 1$ for $n \ge N$; then $|x_n| \le \max\{|x_1|, \dots, |x_{N-1}|, |x| + 1\}$. ■

Theorem 1.14 (1.3.3, limit laws)

If $x_n \to x$ and $y_n \to y$, then (a) $x_n + y_n \to x + y$;
(b) $ax_n \to ax$;
(c) $x_n y_n \to xy$;
(d) if $y \ne 0$, then $y_n \ne 0$ for large n and $x_n/y_n \to x/y$;
(e) if $x_n \le y_n$ for all large n, then $x \le y$;
(f) (squeeze) if $x_n \le z_n \le y_n$ and $x = y$, then $z_n \to x$.

Proof. (a) $|(x_n + y_n) - (x + y)| \le |x_n - x| + |y_n - y|$; use $\epsilon/2$ for each.

(b) is a special case of (c).

(c) With M bounding $|x_n|$ (Proposition 1.13), $|x_n y_n - xy| \le |x_n||y_n - y| + |y||x_n - x| \le M|y_n - y| + |y||x_n - x|$.

(d) For large n, $|y_n| > |y|/2$, and $|1/y_n - 1/y| = |y - y_n|/(|y_n||y|) \le 2|y_n - y|/|y|^2$; combine with (c).

(e) If $x > y$, take $\epsilon = (x-y)/2$; for large n, $x_n > x - \epsilon = y + \epsilon > y_n$.

(f) For large n, $x - \epsilon < x_n \le z_n \le y_n < x + \epsilon$. ■

Example 1.1

(a) $1/n \to 0$ by Corollary 1.6.
(b) For $|r| < 1$, $r^n \to 0$: write $1/|r| = 1 + h$, $h > 0$; then $(1+h)^n \ge 1 + nh$ (Bernoulli), so $|r|^n \le 1/(1+nh)$.
(c) $n^{1/n} \to 1$: with $n^{1/n} = 1 + h_n$, $n \ge \binom{n}{2} h_n^2$, so $0 \le h_n \le \sqrt{2/(n-1)}$.

1.4 Monotone sequences, limsup and liminf

Theorem 1.15 (1.4.1, monotone convergence)

A bounded monotone sequence of real numbers converges.

Proof. Let (x_n) be increasing and bounded, and $x = \sup\{x_n\}$ (axiom (C)). Given ϵ, some $x_N > x - \epsilon$, and then $x - \epsilon < x_N \le x_n \le x$ for $n \ge N$. For decreasing sequences use inf. ■

Corollary 1.16 (1.4.2)

A series of nonnegative terms whose partial sums are bounded converges.

Proof. The partial sums form a bounded increasing sequence. ■

Example 1.2

$(1+1/n)^n$ is increasing and bounded by 3 (binomial expansion), so it converges to a number e. $\sum 1/n^p$ converges for $p > 1$: grouping terms in blocks $2^k \le n < 2^{k+1}$, the partial sums are at most $\sum_k 2^{k(1-p)} < \infty$. $\sum 1/n$ diverges: the block sums are at least $\frac{1}{2}$.

For a bounded sequence (x_n), the sequences $\sup_{k \ge n} x_k$ and $\inf_{k \ge n} x_k$ are decreasing and increasing, so they converge (Theorem 1.15); define

$$\limsup_n x_n = \lim_n \sup_{k \ge n} x_k = \inf_n \sup_{k \ge n} x_k, \qquad \liminf_n x_n = \lim_n \inf_{k \ge n} x_k = \sup_n \inf_{k \ge n} x_k.$$

Clearly $\liminf x_n \le \limsup x_n$, and $\liminf x_n = -\limsup(-x_n)$.

Proposition 1.17 (1.4.3)

For bounded (x_n), $a = \limsup x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n < a + \epsilon$ for all $n \ge N$, and (b) for every $\epsilon > 0$, $x_n > a - \epsilon$ for infinitely many n.

Proof. Let $s_n = \sup_{k \ge n} x_k \downarrow a$.

(a) Choose N with $s_N < a + \epsilon$; then $x_n \le s_N$ for $n \ge N$.

(b) For each N, $s_N \ge a > a - \epsilon$, so some $k \ge N$ has $x_k > a - \epsilon$. Conversely, if a satisfies (a) and (b), then by (a) $s_n \le a + \epsilon$ for large n, so $\limsup \le a + \epsilon$; by (b) $s_n > a - \epsilon$ for all n, so $\limsup \ge a - \epsilon$. As ϵ is arbitrary, $\limsup x_n = a$. ■

Proposition 1.18 (1.4.4)

For bounded (x_n), $a = \liminf x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n > a - \epsilon$ for all $n \geq N$, and (b) for every $\epsilon > 0$, $x_n < a + \epsilon$ for infinitely many n.

Proof. Apply Proposition 1.17 to $(-x_n)$. ■

Theorem 1.19 (1.4.5)

If $x_n \to a$, then $\limsup x_n = \liminf x_n = a$. Conversely, if (x_n) is bounded and $\limsup x_n = \liminf x_n = a$, then $x_n \to a$.

Proof. If $x_n \to a$, then a satisfies conditions (a) and (b) of both propositions. Conversely, part (a) of Propositions 1.17 and 1.18 give $a - \epsilon < x_n < a + \epsilon$ for all large n. ■

Theorem 1.20 (1.4.6, nested interval theorem)

If $I_n = [a_n, b_n]$ are bounded closed intervals with $I_{n+1} \subset I_n$ for all n, then $\bigcap_n I_n \neq \emptyset$. If moreover $b_n - a_n \to 0$, the intersection is a single point.

Proof. (a_n) is increasing and bounded by b_1, so $a_n \to a = \sup a_n$; for all m, n, $a_n \leq a_{\max(m,n)} \leq b_{\max(m,n)} \leq b_m$, so $a \leq b_m$. Thus $a_m \leq a \leq b_m$ for all m. If x, y both lie in every I_n, $|x - y| \leq b_n - a_n \to 0$. ■

Conversely, the nested interval theorem together with the Archimedean property implies the completeness axiom: for S bounded above, bisect an interval $[s_0, u_0]$ ($s_0 \in S$, u_0 an upper bound) repeatedly, keeping the half whose right end point is an upper bound and whose interval meets S; the common point of the intervals is $\sup S$. Similarly, Theorem 1.15 is equivalent to (C).

1.5 Countable and uncountable sets

A set is **countable** if it is finite or in bijection with $\mathbb{N}$.

Proposition 1.21 (countable unions)

Subsets of countable sets are countable; $\mathbb{N} \times \mathbb{N}$ is countable; a countable union of countable sets is countable. Hence $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof. An infinite $A \subset \mathbb{N}$ is enumerated increasingly by recursion ($a_1 = \min A$, $a_{k+1} = \min(A - \{a_1, \ldots, a_k\})$). $(m, n) \mapsto 2^m 3^n$ injects $\mathbb{N} \times \mathbb{N}$ into $\mathbb{N}$. If A_k are countable with surjections $f_k : \mathbb{N} \to A_k$ (for nonempty A_k), then $(k, n) \mapsto f_k(n)$ maps $\mathbb{N} \times \mathbb{N}$ onto $\bigcup A_k$, and a set that is the image of a countable set is countable (choose least preimages). $\mathbb{Q}$ is the image of $\mathbb{Z} \times \mathbb{N}$ under $(p, q) \mapsto p/q$. ■

Theorem 1.22 (Cantor)

$\mathbb{R}$ is uncountable.

Proof. Let $x_1, x_2, \ldots$ be any sequence of reals. Choose a closed interval I_1 of positive length not containing x_1, then $I_2 \subset I_1$ not containing x_2, and so on. By Theorem 1.20 some $x \in \bigcap I_n$, and $x \neq x_n$ for every n. ■

Consequently the irrational numbers are uncountable, and every interval of positive length is uncountable.

Chapter 2. Topology of Euclidean Space

2.1 Euclidean space

$\mathbb{R}^n$ is the set of n-tuples $x = (x_1, \dots, x_n)$ with coordinatewise addition and scalar multiplication, the **inner product** $\langle x, y\rangle = \sum x_i y_i$, the **norm** $\|x\| = \langle x, x\rangle^{1/2}$ and the **distance** $\|x - y\|$.

Theorem 2.1 (2.1.1, Cauchy–Schwarz inequality)

$|\langle x, y\rangle| \le \|x\|\|y\|$ for all $x, y \in \mathbb{R}^n$, with equality iff x, y are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $\|x - z\| \le \|x - y\| + \|y - z\|$.

Proof. If $y = 0$ both sides vanish. Otherwise, for real t, $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$; take $t = \langle x, y\rangle / \|y\|^2$ to get $0 \le \|x\|^2 - \langle x, y\rangle^2/\|y\|^2$. Equality forces $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Also $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, so a sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges.

2.2 Open and closed sets

The **ϵ-neighbourhood** of x is $N(x, \epsilon) = \{y : \|y - x\| < \epsilon\}$. A point x is an **interior point** of A if some $N(x, \epsilon) \subset A$; A is **open** if all its points are interior. x is a **limit point** of A if every $N(x, \epsilon)$ contains a point of A other than x; A' denotes the set of limit points, and $\overline{A} = A \cup A'$ is the **closure**. A is **closed** if $\mathbb{R}^n - A$ is open. Points of $A - A'$ are **isolated points**. Every $N(x, \epsilon)$ is open, by the triangle inequality.

Proposition 2.2 (2.2.1)

(a) $\emptyset$ and $\mathbb{R}^n$ are open;
(b) unions of open sets are open;
(c) finite intersections of open sets are open. Dually, $\emptyset, \mathbb{R}^n$ are closed, intersections of closed sets are closed, and finite unions of closed sets are closed.

Proof. (a), (b) are immediate.

(c) If $x \in U_1 \cap \cdots \cap U_k$ with $N(x, \epsilon_i) \subset U_i$, then $N(x, \min \epsilon_i)$ lies in the intersection. The dual statements follow by De Morgan's laws. ∎

An infinite intersection of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.

Proposition 2.3 (2.2.2)

For $A \subset \mathbb{R}^n$ and $x \in \mathbb{R}^n$:
(a) $x \in A'$ iff some sequence in $A - \{x\}$ converges to x;
(b) $x \in \overline{A}$ iff some sequence in A converges to x.

Proof. (a) If $x \in A'$, pick $x_k \in A \cap N(x, 1/k)$, $x_k \ne x$. Conversely, if $x_k \to x$ with $x_k \in A - \{x\}$, every $N(x, \epsilon)$ contains some x_k.

(b) If $x \in A$ use the constant sequence; otherwise $x \in A'$ and use (a). Conversely, if $x_k \to x$, $x_k \in A$, either $x = x_k$ for some k or the x_k lie in $A - \{x\}$. ∎

Theorem 2.4 (2.2.3)

For $F \subset \mathbb{R}^n$ the following are equivalent:
(a) F is closed;
(b) $F' \subset F$;
(c) $F = \overline{F}$.

Proof. (a)⇒(b): if $x \notin F$, some $N(x,\epsilon)$ misses F, so $x \notin F'$. (b)⇒(c): $\overline{F} = F \cup F' = F$. (c)⇒(a): if $x \notin F$, then $x \notin F'$, so some $N(x,\epsilon)$ contains no point of F (other than x, which is not in F); thus $\mathbb{R}^n - F$ is open. ■

Corollary 2.5 (2.2.4)

A' is closed for every $A \subset \mathbb{R}^n$; hence $\overline{A}$ is closed, and it is the smallest closed set containing A.

Proof. Let $x \in (A')'$ and $\epsilon > 0$. Choose $y \in A'$ with $0 < \|y - x\| < \epsilon$, and $\delta = \min(\|y - x\|, \epsilon - \|y - x\|) > 0$. Then $N(y,\delta)$ contains a point $z \in A$ with $z \neq y$; $z \in N(x,\epsilon)$, and $z \neq x$ because $\|z - y\| < \|x - y\|$. So $x \in A'$. The same argument shows $(\overline{A})' \subset A'$: if $N(x,\epsilon)$ contains $y \in A \cup A'$ with $y \neq x$, either $y \in A$, or $y \in A'$ and a small neighbourhood of y inside $N(x,\epsilon)$ contains a point of A other than x. So $\overline{A}$ is closed by Theorem 2.4. If $F \supset A$ is closed, $F \supset F' \supset A'$, so $F \supset \overline{A}$. ■

For $X \subset \mathbb{R}^n$, a set $U \subset X$ is **open in** X (relatively open) if for each $x \in U$ there is $\epsilon > 0$ with $N(x,\epsilon) \cap X \subset U$.

Proposition 2.6 (2.2.5)

For $U \subset X \subset \mathbb{R}^n$ the following are equivalent:
(a) U is open in X;
(b) $U = V \cap X$ for some open $V \subset \mathbb{R}^n$.

Proof. (a)⇒(b): for $x \in U$ choose ϵ_x with $N(x,\epsilon_x) \cap X \subset U$, and let $V = \bigcup_x N(x,\epsilon_x)$. (b)⇒(a): if $x \in V \cap X$ and $N(x,\epsilon) \subset V$, then $N(x,\epsilon) \cap X \subset U$. ■

Similarly, $F \subset X$ is closed in X iff $F = E \cap X$ with E closed in $\mathbb{R}^n$.

2.3 The Bolzano–Weierstrass theorem and Cauchy sequences

A **box** is $B = I_1 \times \cdots \times I_n$ with bounded closed intervals I_k.

Lemma 2.7 (2.3.1, nested boxes)

If $B_1 \supset B_2 \supset \cdots$ are boxes, then $\bigcap_m B_m \neq \emptyset$.

Proof. Write $B_m = I_1^m \times \cdots \times I_n^m$. For each k, the intervals I_k^m are nested, so contain a common point x_k (Theorem 1.20); then $(x_1, \ldots, x_n) \in \bigcap B_m$. ■

Theorem 2.8 (2.3.2, Bolzano–Weierstrass)

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point.

Proof. A lies in a box B_1. Divide B_1 into 2^n boxes by bisecting each side; one of them, B_2, contains infinitely many points of A. Repeating, we obtain nested boxes B_m whose sides have length 2^{1-m} times those of B_1, each containing infinitely many points of A. Let $x \in \bigcap B_m$ (Lemma 2.7). Any

$N(x,\epsilon)$ contains B_m for large m (its diameter tends to 0), hence infinitely many points of A; so $x \in A'$. ■

Proposition 2.9 (2.3.3)

If x is a limit point of A, then $A \cap N(x,\epsilon)$ is infinite for every $\epsilon > 0$.

Proof. If $A \cap N(x,\epsilon) - \{x\} = \{a_1, \dots, a_k\}$, then $N(x,\delta)$ with $\delta = \min\|a_i - x\|$ contains no point of A other than x, a contradiction. ■

Theorem 2.10 (2.3.4)

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. If the range $\{x_k\}$ is finite, some value is taken infinitely often, giving a constant subsequence. Otherwise it is a bounded infinite set with a limit point x (Theorem 2.8). Each $N(x,1/j)$ contains infinitely many terms (Proposition 2.9), so we can choose $k_1 < k_2 < \cdots$ with $\|x_{k_j} - x\| < 1/j$. ■

For a bounded real sequence (a_n) let A be the set of limits of its convergent subsequences.

Theorem 2.11 (2.3.5)

$\limsup a_n = \max A$ and $\liminf a_n = \min A$.

Proof. Let $a = \limsup a_n$. By Proposition 1.17(b) we may choose $n_1 < n_2 < \cdots$ with $a_{n_k} > a - 1/k$, and by (a) of the same proposition $a_{n_k} < a + 1/k$ for large k; so $a_{n_k} \to a$ and $a \in A$. If $a_{n_k} \to b$, then $a_{n_k} < a + \epsilon$ for large k, so $b \le a$. The $\liminf$ case is symmetric. ■

A sequence is **Cauchy** if for every $\epsilon > 0$ there is N with $\|x_m - x_n\| < \epsilon$ for all $m, n \ge N$.

Proposition 2.12 (2.3.6)

Every convergent sequence is Cauchy, and every Cauchy sequence is bounded.

Proof. If $x_n \to x$, choose N with $\|x_n - x\| < \epsilon/2$ for $n \ge N$; then $\|x_m - x_n\| < \epsilon$. If (x_n) is Cauchy, choose N for $\epsilon = 1$; then $\|x_n\| \le \max\{\|x_1\|, \dots, \|x_N\|\} + 1$. ■

Lemma 2.13 (2.3.7)

If a subsequence of a Cauchy sequence (x_n) converges to x, then (x_n) converges to x.

Proof. Given ϵ, choose N with $\|x_m - x_n\| < \epsilon/2$ for $m, n \ge N$, and k with $n_k \ge N$ and $\|x_{n_k} - x\| < \epsilon/2$. For $n \ge N$, $\|x_n - x\| \le \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \epsilon$. ■

Theorem 2.14 (2.3.8, Cauchy criterion)

Every Cauchy sequence in $\mathbb{R}^n$ converges.

Proof. It is bounded (Proposition 2.12), so has a convergent subsequence (Theorem 2.10); apply Lemma 2.13. ■

Corollary 2.15 (2.3.9, Cauchy criterion for series)

$\sum a_n$ converges iff for every $\epsilon > 0$ there is N with $|a_{m+1} + \cdots + a_n| < \epsilon$ for all $n > m \ge N$.

Proof. Apply Theorem 2.14 to the partial sums s_n, since $s_n - s_m = a_{m+1} + \cdots + a_n$. ■

In particular, if $\sum a_n$ converges then $a_n \to 0$, and if $\sum |a_n|$ converges (**absolute convergence**) then $\sum a_n$ converges, because $|a_{m+1} + \cdots + a_n| \le |a_{m+1}| + \cdots + |a_n|$.

2.4 Series

Proposition 2.16 (2.4.1, rearrangement of nonnegative series)

Let $a_n \ge 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ a bijection. If $\sum a_n = s$, then $\sum a_{\tau(n)} = s$; if $\sum a_n$ diverges, so does $\sum a_{\tau(n)}$.

Proof. Each partial sum $\sum_{n \le N} a_{\tau(n)}$ is at most $\sum_{n \le M} a_n \le s$ with $M = \max \tau(\{1, \dots, N\})$, so $t = \sum a_{\tau(n)} \le s$ by Corollary 1.16. Applying this to τ^{-1} gives $s \le t$. The divergence statement follows by symmetry. ■

More generally, an absolutely convergent series may be rearranged without changing its sum: split a_n into positive and negative parts $a_n^{\pm} = \max(\pm a_n, 0)$, both summable, and apply the proposition to each. By contrast, a conditionally convergent series can be rearranged to have any sum (Riemann).

Theorem 2.17 (2.4.2, alternating series test)

If $x_1 \ge x_2 \ge \cdots \ge 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and its sum s satisfies $|s - s_n| \le x_{n+1}$.

Proof. $s_{2m} = (x_1 - x_2) + \cdots + (x_{2m-1} - x_{2m})$ increases, $s_{2m+1} = x_1 - (x_2 - x_3) - \dots$ decreases, and $s_{2m} \le s_{2m+1}$. Both converge (Theorem 1.15), and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$, so they have the same limit s, which lies between consecutive partial sums. ■

Theorem 2.18 (2.4.3, comparison test)

If $0 \le a_n \le b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.

Proof. The partial sums of $\sum a_n$ are bounded by $\sum b_n$; use Corollary 1.16. ■

Proposition 2.19 (2.4.4, root test)

Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, $\sum a_n$ diverges.

Proof. If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large n (Proposition 1.17), and $\sum \beta^n < \infty$. If $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$. ■

Proposition 2.20 (2.4.5, ratio test)

Let $a_n \ne 0$. If $\limsup |a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely; if $|a_{n+1}/a_n| \ge 1$ for all large n, then $\sum a_n$ diverges.

Proof. If $|a_{n+1}/a_n| \le \beta < 1$ for $n \ge N$, then $|a_n| \le |a_N| \beta^{n-N}$. In the second case $|a_n|$ is eventually nondecreasing and positive, so $a_n \not\to 0$. ■

Proposition 2.21 (2.4.6)

If $c_n > 0$, then $\liminf \frac{c_{n+1}}{c_n} \le \liminf c_n^{1/n} \le \limsup c_n^{1/n} \le \limsup \frac{c_{n+1}}{c_n}$.

Proof. Let $\beta = \limsup c_{n+1}/c_n$, assumed finite, and $\gamma > \beta$. For $n \geq N$, $c_{n+1} < \gamma c_n$, so $c_n < c_N \gamma^{-N} \gamma^n$ and $c_n^{1/n} < (c_N \gamma^{-N})^{1/n} \gamma \to \gamma$. Hence $\limsup c_n^{1/n} \leq \gamma$ for all $\gamma > \beta$. The left inequality is analogous. ■

So the root test is stronger than the ratio test: whenever the ratio test decides, so does the root test.

2.5 Compact sets

An **open cover** of K is a family of open sets whose union contains K; K is **compact** if every open cover has a finite subcover.

Proposition 2.22 (2.5.1)

Every compact set is bounded and closed.

Proof. The sets $N(0,k)$ cover K; finitely many do, so $K \subset N(0,m)$. If $x \notin K$, the open sets $U_k = \{y : \|y - x\| > 1/k\}$ cover K; a finite subcover gives $K \subset U_m$, so $N(x, 1/m)$ misses K. Hence $\mathbb{R}^n - K$ is open. ■

Theorem 2.23 (2.5.2)

Every box $B = I_1 \times \cdots \times I_n$ is compact.

Proof. Suppose an open cover $\{U_i\}$ of B has no finite subcover. Bisecting all sides, one of the 2^n sub-boxes, B_2, is not covered by finitely many U_i; continuing gives nested boxes B_m of diameter $\operatorname{diam} B_1/2^{m-1}$ none finitely covered. Let $x \in \bigcap B_m$ (Lemma 2.7) and $x \in U_i$, $N(x,\epsilon) \subset U_i$. For large m, $B_m \subset N(x,\epsilon) \subset U_i$, a contradiction. ■

Proposition 2.24 (2.5.3)

A closed subset F of a compact set K is compact.

Proof. If $\{U_i\}$ covers F, then $\{U_i\} \cup \{\mathbb{R}^n - F\}$ covers K; a finite subcover, with $\mathbb{R}^n - F$ removed, covers F. ■

Theorem 2.25 (2.5.4, Heine–Borel)

For $K \subset \mathbb{R}^n$ the following are equivalent:
(a) K is compact;
(b) K is bounded and closed;
(c) every infinite subset of K has a limit point in K.

Proof. (a)⇒(b): Proposition 2.22. (b)⇒(a): K lies in a box, which is compact (Theorem 2.23); apply Proposition 2.24. (b)⇒(c): Bolzano–Weierstrass gives a limit point, which is in K as K is closed. (c)⇒(b): if K were unbounded, choose $x_k \in K$ with $\|x_k\| > k$; the infinite set $\{x_k\}$ has no limit point at all. If $x \in K'$, Proposition 2.9 gives distinct $x_k \in K$ with $x_k \to x$; the set $\{x_k\}$ has x as its only limit point, so $x \in K$ by (c), and K is closed. ■

So K is compact iff every sequence in K has a subsequence converging in K.

Theorem 2.26 (2.5.5, finite intersection property)

If $\{K_i : i \in I\}$ are compact sets such that $\bigcap_{i \in J} K_i \neq \emptyset$ for every finite $J \subset I$, then $\bigcap_{i \in I} K_i \neq \emptyset$.

Proof. Fix i_0. If the intersection were empty, the open sets $\mathbb{R}^n - K_i$ would cover K_{i_0}, and finitely many $\mathbb{R}^n - K_{i_1}, \dots, \mathbb{R}^n - K_{i_m}$ would cover it; then $K_{i_0} \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

In particular a decreasing sequence of nonempty compact sets has nonempty intersection.

Corollary 2.27 (2.5.6)

A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point.

Proof. Let $S = \{s_1, s_2, \dots\}$ and suppose no point is isolated. Choose a closed ball $K_1 = \overline{N(y_1, r_1)}$ with $y_1 \in S$; inductively, since y_k is not isolated, $N(y_k, r_k)$ contains infinitely many points of S, so we can choose $y_{k+1} \in S \cap N(y_k, r_k)$, $y_{k+1} \neq s_k$, and a closed ball K_{k+1} about y_{k+1} inside $N(y_k, r_k)$ not containing s_k. The sets $K_k \cap S$ are nonempty, compact and decreasing, so they have a common point (Theorem 2.26); it lies in S but differs from every s_k, a contradiction. ■

Thus a nonempty closed set without isolated points (a **perfect set**), such as the Cantor set, is uncountable.

2.6 Connected sets

$S \subset \mathbb{R}^n$ is **disconnected** if there are open U, V with

$$U \cap S \neq \emptyset, \quad V \cap S \neq \emptyset, \quad U \cap V \cap S = \emptyset, \quad S \subset U \cup V,$$

and **connected** otherwise. A set $S \subset \mathbb{R}$ is an **interval** if $x, y \in S$ and $x < z < y$ imply $z \in S$.

Proposition 2.28 (2.6.1)

A subset S of $\mathbb{R}$ is connected iff it is an interval.

Proof. If S is not an interval, take $x < z < y$ with $x, y \in S$, $z \notin S$; then $U = (-\infty, z)$, $V = (z, \infty)$ disconnect S. Conversely let S be an interval and U, V as above, with $x \in U \cap S$, $y \in V \cap S$, say $x < y$; then $[x, y] \subset S$. Let $c = \sup([x, y] \cap U)$. If $c \in U$, then $c < y$ (as $y \notin U$) and U contains $[c, c + \delta)$, contradicting the definition of c; if $c \in V$, then $c > x$ and $(c - \delta, c] \subset V$, so $(c - \delta, c]$ misses U and c is not the supremum. But $c \in [x, y] \subset S \subset U \cup V$, a contradiction. ■

Proposition 2.29 (2.6.2)

If $\{C_i\}$ are connected and $\bigcap C_i \neq \emptyset$, then $C = \bigcup C_i$ is connected.

Proof. Suppose U, V disconnect C, and let $p \in \bigcap C_i$, say $p \in U$. Each C_i is connected and meets U (at p), so $C_i \cap V = \emptyset$ (otherwise U, V would disconnect C_i). Then $C \cap V = \emptyset$, a contradiction. ■

Convex sets, in particular balls and boxes, are connected: they are unions of segments through a common point, each segment being connected as a continuous image of an interval (Theorem 3.13).

Chapter 3. Continuous Functions

Throughout, $X \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^m$, and functions $f : X \to Y$.

3.1 Limits and continuity

Let x_0 be a limit point of X. We write $\lim_{x\to x_0} f(x) = y_0$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - y_0\| < \epsilon$ whenever $x \in X$ and $0 < \|x - x_0\| < \delta$. f is **continuous at** $x_0 \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(x_0)\| < \epsilon$ whenever $x \in X$, $\|x - x_0\| < \delta$; equivalently, x_0 is isolated in X, or $\lim_{x\to x_0} f(x) = f(x_0)$. f is **continuous** if it is continuous at every point.

Proposition 3.1 (3.1.1)

If $\lim_{x\to x_0} f(x) = y_0$ and $\lim_{x\to x_0} g(x) = z_0$, then $f + g \to y_0 + z_0$, $af \to ay_0$, and for real-valued functions $fg \to y_0 z_0$, and $f/g \to y_0/z_0$ if $z_0 \neq 0$.

Proof. As in Theorem 1.14, or by Proposition 3.3 below applied to sequences. ■

Proposition 3.2 (3.1.2)

If $f, g : X \to \mathbb{R}^m$ are continuous, so is $af + bg$; for real-valued f, g, so are fg and f/g (where $g \neq 0$).

Proof. Apply Proposition 3.1 at each non-isolated point. ■

Polynomials in several variables are therefore continuous, and so are rational functions where the denominator does not vanish.

Proposition 3.3 (3.1.3, sequential criterion)

f is continuous at x_0 iff $f(x_k) \to f(x_0)$ for every sequence $x_k \to x_0$ in X.

Proof. ($\Rightarrow$) Given ϵ, take δ; for large k, $\|x_k - x_0\| < \delta$, so $\|f(x_k) - f(x_0)\| < \epsilon$.

($\Leftarrow$) If f is not continuous at x_0, for some ϵ and each k there is x_k with $\|x_k - x_0\| < 1/k$ but $\|f(x_k) - f(x_0)\| \geq \epsilon$. ■

Similarly $\lim_{x\to x_0} f(x) = y_0$ iff $f(x_k) \to y_0$ for every sequence $x_k \to x_0$ with $x_k \neq x_0$.

Proposition 3.4 (3.1.4)

If $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.

Proof. Given ϵ, choose η for g at $f(x_0)$ and then δ for f at x_0 with η in place of ϵ. ■

Example 3.1

(a) $f(x) = \sin(1/x)$ $(x \neq 0)$ has no limit at 0: $x_k = 1/(k\pi + \pi/2)$ gives ± 1 alternately.
(b) $x \sin(1/x) \to 0$ by the squeeze $|x \sin(1/x)| \leq |x|$.
(c) Thomae's function ($f(p/q) = 1/q$ in lowest terms, $f = 0$ at irrationals) is continuous exactly at the irrationals.
(d) $f(x, y) = xy/(x^2 + y^2)$, $f(0, 0) = 0$, is continuous along every line through the origin, but $f(t, t) = \frac{1}{2}$, so f is not continuous at 0.

Theorem 3.5 (3.1.5)

For $f : X \to Y$ the following are equivalent:
(a) f is continuous;
(b) $f^{-1}(V)$ is open in X for every open V in Y;
(c) $f^{-1}(E)$ is closed in X for every closed E in Y.

Proof. (a)⇒(b): if $x \in f^{-1}(V)$ and $N(f(x), \epsilon) \cap Y \subset V$, the δ of continuity gives $N(x, \delta) \cap X \subset f^{-1}(V)$. (b)⇒(a): $f^{-1}(N(f(x_0), \epsilon) \cap Y)$ is open in X and contains x_0. (b)⇔(c): $f^{-1}(Y - E) = X - f^{-1}(E)$. ■

3.2 Continuity, compactness and connectedness

Theorem 3.6 (3.2.1)

If $f : X \to Y$ is continuous and surjective and X is compact, then Y is compact.

Proof. Let $\{U_i\}$ be an open cover of Y; by Theorem 3.5 and Proposition 2.6, $f^{-1}(U_i) = V_i \cap X$ with V_i open in $\mathbb{R}^n$, and the V_i cover X. Finitely many $V_{i_1}, \ldots, V_{i_k}$ do; then $U_{i_1}, \ldots, U_{i_k}$ cover $Y = f(X)$. ■

Corollary 3.7 (3.2.2, extreme value theorem)

A continuous $f : X \to \mathbb{R}$ on a compact set attains its maximum and minimum.

Proof. $f(X)$ is compact, hence bounded and closed; so $\sup f(X)$ is finite and, being a limit of points of $f(X)$, lies in $f(X)$. ■

Corollary 3.8 (3.2.3)

If X is compact and $f : X \to \mathbb{R}$ is continuous with $f > 0$, then there is $\delta > 0$ with $f(x) \geq \delta$ for all x.

Proof. Take $\delta = \min f > 0$. ■

Corollary 3.9 (3.2.4)

A continuous $f : [a, b] \to \mathbb{R}$ attains its maximum and minimum.

Corollary 3.10 (3.2.5)

If X is compact and $f : X \to Y$ is a continuous bijection, then f^{-1} is continuous.

Proof. Let U be open in X. $X - U$ is closed in the compact X, so compact (Proposition 2.24); then $f(X - U) = Y - f(U)$ is compact, hence closed, so $f(U)$ is open in Y. By Theorem 3.5, f^{-1} is continuous. ■

Compactness is needed: $t \mapsto (\cos t, \sin t)$ is a continuous bijection $[0, 2\pi) \to S^1$ whose inverse is not continuous. For nonempty $A, B \subset \mathbb{R}^n$ let $\operatorname{dist}(A, B) = \inf\{\|a - b\| : a \in A, b \in B\}$.

Proposition 3.11 (3.2.6)

If A is compact, B is closed and $A \cap B = \emptyset$, then $\operatorname{dist}(A, B) > 0$.

Proof. The function $d(x) = \operatorname{dist}(\{x\}, B)$ is continuous (indeed $|d(x) - d(y)| \le \|x - y\|$) and positive on A, since $\mathbb{R}^n - B$ is open. By Corollary 3.8 it is bounded below on A by some $\delta > 0$. ■

The compactness is needed: the closed sets $\{(x, 0)\}$ and $\{(x, 1/x) : x > 0\}$ are at distance 0.

Lemma 3.12 (3.2.7)

S is disconnected iff there is a continuous surjection $f : S \to \{0, 1\}$.

Proof. If U, V disconnect S, let $f = 0$ on $U \cap S$ and 1 on $V \cap S$; preimages of subsets of $\{0, 1\}$ are relatively open, so f is continuous. Conversely, $f^{-1}(0)$ and $f^{-1}(1)$ are nonempty and relatively open (Theorem 3.5), so they are $U \cap S$ and $V \cap S$ with U, V open (Proposition 2.6); then $U \cap V \cap S = f^{-1}(0) \cap f^{-1}(1) = \emptyset$ and $S \subset U \cup V$. ■

Theorem 3.13 (3.2.8)

If $f : X \to Y$ is continuous and surjective and X is connected, then Y is connected.

Proof. If Y were disconnected, Lemma 3.12 gives a continuous surjection $g : Y \to \{0, 1\}$, and then $g \circ f : X \to \{0, 1\}$ is a continuous surjection. ■

Corollary 3.14 (3.2.9)

If $I \subset \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$ is continuous, then $f(I)$ is an interval.

Proof. Proposition 2.28 and Theorem 3.13. ■

Corollary 3.15 (3.2.10, intermediate value theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < \alpha < f(b)$, then $f(c) = \alpha$ for some $c \in (a, b)$.

Corollary 3.16 (3.2.11)

If $f : [a, b] \to \mathbb{R}$ is continuous, then $f([a, b])$ is a bounded closed interval $[m, M]$.

Proof. It is an interval (Corollary 3.14) and compact (Theorem 3.6), containing its minimum m and maximum M. ■

A function mapping intervals to intervals need not be continuous, for example $\sin(1/x)$ extended by 0 at 0.

Corollary 3.17 (3.2.12, fixed point)

Every continuous $f : [a, b] \to [a, b]$ has a fixed point.

Proof. $g(x) = x - f(x)$ has $g(a) \le 0 \le g(b)$; apply Corollary 3.15 (or $g(a) = 0$ or $g(b) = 0$). ■

Example 3.2

Every polynomial of odd degree has a real root (it takes both signs). Every $a > 0$ has a unique positive n-th root, since x^n is continuous, strictly increasing and unbounded on $[0, \infty)$.

3.3 Uniform continuity

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(y)\| < \epsilon$ whenever $x, y \in X$, $\|x - y\| < \delta$.

Theorem 3.18 (3.3.1, Heine)

A continuous function on a compact set is uniformly continuous.

Proof. Given ϵ, choose for each $x \in X$ a $\delta(x) > 0$ with $\|f(y)-f(x)\| < \epsilon/2$ for $y \in X$, $\|y-x\| < \delta(x)$. The sets $N(x, \delta(x)/2)$ cover X; finitely many with centres $x_1, \ldots, x_k$ do. Let $\delta = \min \delta(x_i)/2$. If $\|x - y\| < \delta$, then $\|x - x_i\| < \delta(x_i)/2$ for some i, and $\|y - x_i\| < \delta(x_i)$; so $\|f(x) - f(y)\| \leq \|f(x) - f(x_i)\| + \|f(x_i) - f(y)\| < \epsilon$. ■

Example 3.3

x^2 is not uniformly continuous on $\mathbb{R}$ ($(n+1/n)^2 - n^2 > 2$), and $1/x$ is not uniformly continuous on $(0, 1)$; $\sqrt{x}$ is uniformly continuous on $[0, \infty)$ since $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$.

Theorem 3.19 (3.3.2)

A uniformly continuous function maps Cauchy sequences to Cauchy sequences.

Proof. Given ϵ, take δ from uniform continuity and N with $\|x_m - x_n\| < \delta$ for $m, n \geq N$. ■

Corollary 3.20 (3.3.3)

A uniformly continuous $f : X \to \mathbb{R}^m$ extends uniquely to a continuous function on $\overline{X}$, which is again uniformly continuous.

Proof. For $x_0 \in \overline{X}$ pick $x_k \to x_0$ in X; $(f(x_k))$ is Cauchy (Theorem 3.19), hence converges (Theorem 2.14); define $F(x_0)$ as the limit. Interlacing two such sequences shows the limit does not depend on the sequence, and $F = f$ on X. If $\|x - y\| < \delta$ in $\overline{X}$, approximate by $x_k \to x$, $y_k \to y$ with $\|x_k - y_k\| < \delta$; then $\|F(x) - F(y)\| = \lim \|f(x_k) - f(y_k)\| \leq \epsilon$. Uniqueness follows from Proposition 3.3. ■

3.4 Monotone functions

For $f : (a, b) \to \mathbb{R}$ write $f(x+) = \lim_{t \to x^+} f(t)$ and $f(x-) = \lim_{t \to x^-} f(t)$ when these exist.

Theorem 3.21 (3.4.1)

A monotone $f : (a, b) \to \mathbb{R}$ has one-sided limits at every point, and $f(a+)$, $f(b-)$ exist in $[-\infty, \infty]$. If f is increasing and $a < x < y < b$, then $f(x-) \leq f(x) \leq f(x+) \leq f(y-) \leq f(y) \leq f(y+)$, and $f(x+) = \inf_{t > x} f(t)$, $f(x-) = \sup_{t < x} f(t)$.

Proof. Let f increase. $A = \{f(t) : a < t < x\}$ is bounded above by $f(x)$; let $s = \sup A$. Given ϵ, some $f(t_0) > s - \epsilon$, and for $t_0 < t < x$, $s - \epsilon < f(t_0) \leq f(t) \leq s$; so $f(x-) = s \leq f(x)$. Similarly $f(x+) = \inf_{t > x} f(t) \geq f(x)$. For $x < y$, choosing $x < t < y$, $f(x+) \leq f(t) \leq f(y-)$. ■

Corollary 3.22 (3.4.2)

The set of discontinuities of a monotone function on an interval is countable.

Proof. For increasing f, f is discontinuous at x iff $f(x-) < f(x+)$. Choose a rational $r(x) \in (f(x-), f(x+))$; by Theorem 3.21 these open intervals are disjoint for distinct x, so $x \mapsto r(x)$ is injective into $\mathbb{Q}$. ■

Conversely, for any countable set $\{x_k\} \subset (a,b)$, $f(x) = \sum_{x_k<x} 2^{-k}$ is increasing and discontinuous exactly at the x_k.

Theorem 3.23 (3.4.3)

For a continuous $f : I \to \mathbb{R}$ on an interval, f is injective iff it is strictly monotone. In that case $f(I)$ is an interval and $f^{-1} : f(I) \to I$ is continuous.

Proof. Strictly monotone functions are injective. Conversely, suppose f injective but not strictly monotone; then there are $x < y < z$ in I with $f(y)$ not between $f(x)$ and $f(z)$, say $f(y) > \max(f(x), f(z))$. A value α with $\max(f(x), f(z)) < \alpha < f(y)$ is attained in (x, y) and in (y, z) (Corollary 3.15), contradicting injectivity. If f is strictly increasing, f^{-1} is strictly increasing on the interval $f(I)$ (Corollary 3.14) and maps it onto the interval I; a monotone function with an interval as image has no jumps (a jump $g(y-) < g(y+)$ would omit values by Theorem 3.21), so f^{-1} is continuous. ■

Chapter 4. Differentiable Functions

4.1 Differentiability

Let I be an interval. $f : I \to \mathbb{R}$ is **differentiable** at $x_0 \in I$ if $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists (one-sided at an end point).

Proposition 4.1 (4.1.1)

For $f : (a, b) \to \mathbb{R}$ and $x_0 \in (a, b)$ the following are equivalent:
(a) f is differentiable at x_0;
(b) there are $\lambda \in \mathbb{R}$ and a function ϕ with $f(x_0+h)-f(x_0) = \lambda h+|h|\phi(h)$ and $\lim_{h\to 0} \phi(h) = 0$.
Then $\lambda = f'(x_0)$.

Proof. (a)⇒(b): put $\lambda = f'(x_0)$ and $\phi(h) = (f(x_0 + h) - f(x_0) - \lambda h)/|h|$ for $h \neq 0$, $\phi(0) = 0$. (b)⇒(a): $\frac{f(x_0+h)-f(x_0)}{h} = \lambda \pm \phi(h) \to \lambda$. ■

The linear map $h \mapsto \lambda h$ is the best linear approximation of $f(x_0 + h) - f(x_0)$; this is the form of the derivative that generalizes to several variables.

Proposition 4.2 (4.1.2)

If f is differentiable at x_0, it is continuous at x_0.

Proof. $f(x_0 + h) - f(x_0) = \lambda h + |h|\phi(h) \to 0$. ■

Proposition 4.3 (4.1.3)

If f, g are differentiable at x_0, so are $f+g$, fg, and f/g (if $g(x_0) \neq 0$), with $(f+g)' = f'+g'$, $(fg)' = f'g + fg'$, $(f/g)' = (f'g - fg')/g^2$ at x_0.

Proof. For the product, $\frac{f(x_0+h)g(x_0+h)-f(x_0)g(x_0)}{h} = \frac{f(x_0+h)-f(x_0)}{h} g(x_0+h) + f(x_0)\frac{g(x_0+h)-g(x_0)}{h}$, and $g(x_0 + h) \to g(x_0)$ by Proposition 4.2. For the quotient, first $\frac{1/g(x_0+h)-1/g(x_0)}{h} = -\frac{g(x_0+h)-g(x_0)}{h} \cdot \frac{1}{g(x_0+h)g(x_0)} \to -g'/g^2$, then use the product rule. ■

Proposition 4.4 (4.1.4, chain rule)

If $f : I \to J$ is differentiable at x_0 and $g : J \to \mathbb{R}$ is differentiable at $y_0 = f(x_0)$, then $g \circ f$ is differentiable at x_0 and $(g \circ f)'(x_0) = g'(y_0)f'(x_0)$.

Proof. Write $g(y_0 + k) - g(y_0) = k(g'(y_0) + \psi(k))$ with $\psi(k) \to 0 = \psi(0)$. With $k = f(x_0 + h) - f(x_0) \to 0$,

$$\frac{g(f(x_0 + h)) - g(f(x_0))}{h} = \frac{k}{h}\,(g'(y_0) + \psi(k)) \to f'(x_0)g'(y_0).$$

■

Example 4.1

$f(x) = x^2 \sin(1/x)$, $f(0) = 0$, is differentiable everywhere with $f'(0) = \lim h \sin(1/h) = 0$, but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ is not continuous at 0. $|x|$ is continuous but not differentiable at 0.

4.2 Mean value theorems

If f has a local maximum or minimum at an interior point c and is differentiable there, then $f'(c) = 0$: the difference quotients have opposite signs on the two sides.

Lemma 4.5 (4.2.1, Rolle)

If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

Proof. If f is constant, any c works. Otherwise f attains a maximum or minimum (Corollary 3.9) at a point $c \in (a, b)$ where $f(c) \neq f(a)$, and $f'(c) = 0$. ■

Theorem 4.6 (4.2.2, Cauchy mean value theorem)

If $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b), there is $c \in (a, b)$ with $f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]$.

Proof. Apply Rolle to $h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)]$, for which $h(a) = h(b) = f(a)g(b) - g(a)f(b)$. ■

Corollary 4.7 (4.2.3, mean value theorem)

If f is continuous on $[a, b]$ and differentiable on (a, b), then $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$.

Proof. Take $g(x) = x$. ■

Corollary 4.8 (4.2.4)

For differentiable $f : (a, b) \to \mathbb{R}$: f is increasing iff $f' \geq 0$; constant iff $f' = 0$; decreasing iff $f' \leq 0$. If $f' > 0$, f is strictly increasing.

Proof. If $f' \geq 0$ and $x < y$, $f(y) - f(x) = f'(c)(y - x) \geq 0$. Conversely, if f increases, its difference quotients are ≥ 0, so $f' \geq 0$. ■

Theorem 4.9 (4.2.5, L'Hôpital's rule)

Let f, g be differentiable near x_0 (except possibly at x_0) with $g' \neq 0$ there, $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = 0$, and $\lim_{x\to x_0} f'(x)/g'(x) = L$. Then $\lim_{x\to x_0} f(x)/g(x) = L$. The same holds for one-sided limits, for $x \to \pm\infty$, and when $g(x) \to \pm\infty$.

Proof. Set $f(x_0) = g(x_0) = 0$, making f, g continuous at x_0. For x near x_0, $g(x) \neq 0$ (by Rolle, as $g' \neq 0$), and Theorem 4.6 on the interval between x_0 and x gives $f(x)/g(x) = f'(c_x)/g'(c_x)$ with c_x between x_0 and x; as $x \to x_0$, $c_x \to x_0$. For $x \to \infty$, substitute $x = 1/t$. If $g \to \infty$ as $x \to x_0^+$: given ϵ, choose δ with $|f'/g' - L| < \epsilon$ on $(x_0, x_0 + \delta)$ and fix y there. For $x_0 < x < y$, $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(c)}{g'(c)}$, and

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(y)}{g(x) - g(y)} \cdot \frac{1 - g(y)/g(x)}{1} + \frac{f(y)}{g(x)},$$

where $g(y)/g(x) \to 0$ and $f(y)/g(x) \to 0$ as $x \to x_0^+$; so $\limsup |f/g - L| \leq \epsilon$. ■

Theorem 4.10 (4.2.6, Darboux)

If $F : [a, b] \to \mathbb{R}$ is differentiable and $F'(a) < \alpha < F'(b)$, then $F'(c) = \alpha$ for some $c \in (a, b)$.

Proof. $G(x) = F(x) - \alpha x$ has $G'(a) < 0 < G'(b)$, so $G(x) < G(a)$ for x slightly larger than a and $G(x) < G(b)$ for x slightly smaller than b. Hence the minimum of G on $[a, b]$ is attained at an interior point c, where $G'(c) = 0$. ■

Corollary 4.11 (4.2.7)

If $f : I \to \mathbb{R}$ has a primitive on the interval I, then $f(J)$ is an interval for every interval $J \subset I$.

Proof. Apply Theorem 4.10 to the primitive on subintervals of J. ■

So a function with a jump discontinuity, such as the sign function, has no primitive; derivatives can be discontinuous (Example 4.1) but have the intermediate value property.

4.3 Taylor's theorem

For f with n derivatives at a, let $T_n^f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x-a)^k$ be its Taylor polynomial at a.

Theorem 4.12 (4.3.1, generalized mean value theorem)

Let I be the closed interval with end points $a \neq b$, and $f, g : I \to \mathbb{R}$ with $f^{(n)}, g^{(n)}$ continuous on I and differentiable on $\operatorname{Int} I$. For each $x \in I$, $x \neq a$, there is c strictly between a and x with

$$[f(x) - T_n^f(x)]g^{(n+1)}(c) = [g(x) - T_n^g(x)]f^{(n+1)}(c).$$

Proof. Fix x and let $F(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x-t)^k$, $G(t)$ likewise for g, for t between a and x. In $F'(t)$ the terms telescope:

$$F'(t) = \sum_{k=0}^{n} \frac{f^{(k+1)}(t)}{k!}(x-t)^k - \sum_{k=1}^{n} \frac{f^{(k)}(t)}{(k-1)!}(x-t)^{k-1} = \frac{f^{(n+1)}(t)}{n!}(x-t)^n,$$

and similarly for G. Also $F(x) = f(x)$ and $F(a) = T_n^f(x)$. Theorem 4.6 gives c with $[F(x) - F(a)]G'(c) = [G(x) - G(a)]F'(c)$; cancel the nonzero factor $(x-c)^n/n!$. ■

Corollary 4.13 (4.3.2, Taylor's theorem)

Under the hypotheses on f, for each $x \in I$ there is c between a and x with

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Apply Theorem 4.12 with $g(t) = (t-a)^{n+1}$: $T_n^g = 0$ and $g^{(n+1)} = (n+1)!$. ■

The last term is the **remainder**. If f is C^∞, the **Taylor series** of f at a is $\sum_k \frac{f^{(k)}(a)}{k!}(x-a)^k$; it may fail to converge to f: $f(x) = e^{-1/x^2}$, $f(0) = 0$, has all derivatives 0 at 0.

Proposition 4.14 (4.3.3)

Let f be C^∞ on I, $a, x \in I$, and J the closed interval between them. If $\{|f^{(n)}(t)| : t \in J, n \geq 0\}$ is bounded, the Taylor series of f at a converges to $f(x)$.

Proof. With M a bound, the remainder is at most $M|x-a|^{n+1}/(n+1)! \to 0$ (the terms of the convergent series $\sum r^n/n!$ tend to 0). ■

Example 4.2

e^x, $\sin x$, $\cos x$ satisfy the hypothesis on every bounded interval, so $e^x = \sum x^n/n!$, $\sin x = \sum(-1)^n x^{2n+1}/(2n+1)!$, $\cos x = \sum(-1)^n x^{2n}/(2n)!$ for all real x. Taylor's theorem also gives the second derivative test: if $f'(a) = 0$ and $f''(a) > 0$, f has a strict local minimum at a.

Chapter 5. The Riemann–Stieltjes Integral

5.1 The Riemann integral

A **partition** of $[a,b]$ is a finite set $P=\{a=x_0<x_1<\cdots<x_n=b\}$; write $\Delta x_i=x_i-x_{i-1}$ and $\|P\|=\max\Delta x_i$ (the mesh). $Q\supset P$ is a **refinement** of P. For bounded $f:[a,b]\to\mathbb{R}$ let $M_i=\sup_{[x_{i-1},x_i]}f$, $m_i=\inf_{[x_{i-1},x_i]}f$, and

$$U(f,P)=\sum M_i\Delta x_i,\qquad L(f,P)=\sum m_i\Delta x_i.$$

The **upper** and **lower integrals** are $\overline{\int}f=\inf_P U(f,P)$ and $\underline{\int}f=\sup_P L(f,P)$; f is **Riemann integrable** ($f\in\mathcal{R}$) if they are equal, and the common value is $\int_a^b f$.

Proposition 5.1 (5.1.1)

If $P\subset Q$, then $L(f,P)\le L(f,Q)\le U(f,Q)\le U(f,P)$. For any partitions P,P', $L(f,P)\le U(f,P')$; hence $\underline{\int}f\le\overline{\int}f$.

Proof. It suffices to add one point $y\in(x_{i-1},x_i)$: the infima of f over $[x_{i-1},y]$ and $[y,x_i]$ are $\ge m_i$, so L can only increase; similarly U decreases. Then $L(f,P)\le L(f,P\cup P')\le U(f,P\cup P')\le U(f,P')$. ■

Proposition 5.2 (5.1.2, Riemann's criterion)

A bounded f is integrable iff for every $\epsilon>0$ there is a partition P with $U(f,P)-L(f,P)<\epsilon$.

Proof. If the condition holds, $0\le\overline{\int}f-\underline{\int}f\le U(f,P)-L(f,P)<\epsilon$. Conversely, if $f\in\mathcal{R}$, choose P_1,P_2 with $U(f,P_1)<\int f+\epsilon/2$ and $L(f,P_2)>\int f-\epsilon/2$; then $P=P_1\cup P_2$ works by Proposition 5.1. ■

Theorem 5.3 (5.1.3)

If $f,g\in\mathcal{R}$ and $c\in\mathbb{R}$, then $f+g, cf\in\mathcal{R}$ with $\int(f+g)=\int f+\int g$ and $\int cf=c\int f$; and $f\le g$ implies $\int f\le\int g$.

Proof. On each subinterval $\sup(f+g)\le\sup f+\sup g$ and $\inf(f+g)\ge\inf f+\inf g$, so

$$L(f,P)+L(g,P)\le L(f+g,P)\le U(f+g,P)\le U(f,P)+U(g,P).$$

Choosing P with $U-L<\epsilon/2$ for both f and g (common refinement) gives integrability and the formula. For $c\ge0$, $U(cf,P)=cU(f,P)$ and $L(cf,P)=cL(f,P)$; for $c=-1$, $U(-f,P)=-L(f,P)$. If $f\le g$, then $U(f,P)\le U(g,P)$ for all P. ■

Products of integrable functions are integrable: $f^2\in\mathcal{R}$ because on each subinterval $\sup f^2-\inf f^2\le 2K(M_i-m_i)$ with $K=\sup|f|$, and $fg=\frac14[(f+g)^2-(f-g)^2]$.

Theorem 5.4 (5.1.4)

If $f\in\mathcal{R}[a,b]$ and $a<c<b$, then $f\in\mathcal{R}[a,c]$, $f\in\mathcal{R}[c,b]$ and $\int_a^b f=\int_a^c f+\int_c^b f$. Conversely, integrability on $[a,c]$ and $[c,b]$ implies integrability on $[a,b]$.

Proof. For partitions P of $[a,b]$ containing c, with $P_1=P\cap[a,c]$ and $P_2=P\cap[c,b]$, we have $U(f,P)=U(f,P_1)+U(f,P_2)$, likewise for L. If $U(f,P)-L(f,P)<\epsilon$ (we may add c to P), the

same holds for P_1 and P_2. Conversely, combine partitions of the two halves. Taking infima of upper sums gives the formula. ■

With the convention $\int_b^a f = -\int_a^b f$, the formula holds for any order of a, b, c.

Theorem 5.5 (5.1.5)

If $f \in \mathcal{R}$, then $|f| \in \mathcal{R}$ and $\left|\int_a^b f\right| \le \int_a^b |f|$.

Proof. From $\big||f(x)| - |f(y)|\big| \le |f(x) - f(y)|$, the oscillation of $|f|$ on each subinterval is at most that of f, so $U(|f|, P) - L(|f|, P) \le U(f, P) - L(f, P)$. Then $-|f| \le f \le |f|$ and Theorem 5.3 give the inequality. ■

Example 5.1

The Dirichlet function (1 on $\mathbb{Q}$, 0 elsewhere) is not integrable on $[0, 1]$: every $U(f, P) = 1$ and every $L(f, P) = 0$. Integrability of $|f|$ does not imply that of f: take $f = 1$ on $\mathbb{Q}$ and -1 elsewhere.

5.2 Integrable functions

Theorem 5.6 (5.2.1)

Every continuous $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. f is uniformly continuous (Theorem 3.18): choose δ with $|f(x) - f(y)| < \epsilon/(b - a)$ when $|x - y| < \delta$. If $\|P\| < \delta$, then $M_i - m_i < \epsilon/(b - a)$ (the extrema are attained), so $U(f, P) - L(f, P) < \epsilon$. ■

Theorem 5.7 (5.2.2)

Every monotone $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. Let f increase and P divide $[a, b]$ into n equal parts. Then $M_i = f(x_i)$, $m_i = f(x_{i-1})$, and $U(f, P) - L(f, P) = \frac{b-a}{n} \sum (f(x_i) - f(x_{i-1})) = \frac{b-a}{n}(f(b) - f(a)) < \epsilon$ for large n. ■

More generally, a bounded function with finitely many discontinuities is integrable (enclose them in short intervals), and so is one whose discontinuity set has measure zero (Chapter 10 of the second volume).

A **Riemann sum** is $S(f, P) = \sum f(t_i)\Delta x_i$ with $t_i \in [x_{i-1}, x_i]$.

Theorem 5.8 (5.2.3)

For bounded f and $A \in \mathbb{R}$ the following are equivalent:
(a) $f \in \mathcal{R}$ and $\int_a^b f = A$;
(b) for every $\epsilon > 0$ there is $\delta > 0$ with $|S(f, P) - A| < \epsilon$ for every partition with $\|P\| < \delta$ and every choice of the t_i.

Proof. (a)⇒(b): let $K = \sup|f|$ and choose P_0, with k points, such that $U(f, P_0) - L(f, P_0) < \epsilon/2$. Let $\|P\| < \delta = \epsilon/(8Kk)$. Passing from P to $P \cup P_0$ changes only the at most k subintervals of P that contain points of P_0 in their interior, and each change of U or L is at most $2K\|P\|$. So $U(f, P) - L(f, P) \le U(f, P \cup P_0) - L(f, P \cup P_0) + 4Kk\delta < \epsilon$. Since both $S(f, P)$ and A lie in $[L(f, P), U(f, P)]$, $|S(f, P) - A| < \epsilon$.

(b)⇒(a): fix P with $\|P\| < \delta$. Choosing the t_i with $f(t_i) > M_i - \epsilon/(b-a)$ gives $U(f,P) < S(f,P) + \epsilon < A + 2\epsilon$; similarly $L(f,P) > A - 2\epsilon$. So $U(f,P) - L(f,P) < 4\epsilon$, and $\int f = A$. ∎

Example 5.2

$\int_0^1 x^2\,dx = \lim_n \frac{1}{n}\sum_{k=1}^n (k/n)^2 = \lim \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}$. Also $\lim_n \sum_{k=1}^n \frac{1}{n+k} = \int_0^1 \frac{dx}{1+x} = \log 2$.

5.3 The fundamental theorem of calculus

For $f \in \mathcal{R}[a,b]$ let $F(x) = \int_a^x f$.

Theorem 5.9 (5.3.1)

F is uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 and $F'(x_0) = f(x_0)$.

Proof. With $K = \sup|f|$, $|F(y) - F(x)| = |\int_x^y f| \le K|y-x|$. If f is continuous at x_0, given ϵ choose δ with $|f(t) - f(x_0)| < \epsilon$ for $|t - x_0| < \delta$; then for $0 < |h| < \delta$,

$$\left|\frac{F(x_0+h) - F(x_0)}{h} - f(x_0)\right| = \left|\frac{1}{h}\int_{x_0}^{x_0+h} (f(t) - f(x_0))\,dt\right| \le \epsilon.$$

∎

Corollary 5.10 (5.3.2)

Every continuous $f : [a,b] \to \mathbb{R}$ has a primitive, namely $F(x) = \int_a^x f$.

Theorem 5.11 (5.3.3, fundamental theorem of calculus)

If $F : [a,b] \to \mathbb{R}$ is differentiable and $f = F'$ is integrable, then $\int_a^b f = F(b) - F(a)$.

Proof. For any partition, the mean value theorem gives $t_i \in (x_{i-1}, x_i)$ with $F(x_i) - F(x_{i-1}) = f(t_i)\Delta x_i$, so $F(b) - F(a) = \sum f(t_i)\Delta x_i \in [L(f,P), U(f,P)]$. As $f \in \mathcal{R}$, $F(b) - F(a) = \int f$. ∎

The integrability hypothesis cannot be dropped: there are differentiable functions with bounded, non-integrable derivatives (Volterra), and $x^2 \sin(1/x^2)$ has an unbounded derivative on $[0,1]$.

Proposition 5.12 (5.3.4, change of variables)

If $g : [a,b] \to [c,d]$ is a surjective C^1 function and $f : [c,d] \to \mathbb{R}$ is continuous, then $\int_{g(a)}^{g(b)} f(x)\,dx = \int_a^b f(g(t))g'(t)\,dt$.

Proof. Let F be a primitive of f (Corollary 5.10). By the chain rule $(F \circ g)' = (f \circ g)g'$, which is continuous, hence integrable, so by Theorem 5.11 the right side is $F(g(b)) - F(g(a))$, which is also the left side. ∎

Proposition 5.13 (5.3.5, integration by parts)

If $F, G : [a,b] \to \mathbb{R}$ are differentiable and $f = F'$, $g = G'$ are integrable, then $\int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG$.

Proof. F, G are continuous, hence integrable, so $(FG)' = fG + Fg$ is integrable (products of integrable functions); apply Theorem 5.11 to FG. ∎

5.4 Functions of bounded variation

For $f:[a,b]\to\mathbb{R}$ and a partition P, let $V(f,P)=\sum|f(x_i)-f(x_{i-1})|$. The **total variation** is $V_a^b(f)=\sup_P V(f,P)$; f is of **bounded variation** if it is finite. Refining P increases $V(f,P)$ (triangle inequality). Monotone functions have $V_a^b(f)=|f(b)-f(a)|$; Lipschitz functions with constant K have $V_a^b(f)\le K(b-a)$; BV functions are bounded, $|f(x)|\le|f(a)|+V_a^b(f)$.

Proposition 5.14 (5.4.1)

If f,g are of bounded variation, then so are $f+g$ and fg, with

$$V(f+g)\le V(f)+V(g),\qquad V(fg)\le\|f\|_\infty V(g)+\|g\|_\infty V(f).$$

If moreover $|f|\ge c>0$, then $1/f$ is of bounded variation with $V(1/f)\le V(f)/c^2$.

Proof. Termwise, $|(f+g)(x_i)-(f+g)(x_{i-1})|\le|\Delta f_i|+|\Delta g_i|$, and

$$|f(x_i)g(x_i)-f(x_{i-1})g(x_{i-1})|\le|f(x_i)||\Delta g_i|+|g(x_{i-1})||\Delta f_i|,$$

and $|1/f(x_i)-1/f(x_{i-1})|=|\Delta f_i|/|f(x_i)f(x_{i-1})|\le|\Delta f_i|/c^2$. Sum and take suprema. ∎

Theorem 5.15 (5.4.2)

If f is of bounded variation on $[a,b]$ and $a<c<b$, then f is of bounded variation on $[a,c]$ and $[c,b]$, and $V_a^b(f)=V_a^c(f)+V_c^b(f)$. Conversely, bounded variation on both halves implies it on $[a,b]$.

Proof. For partitions $P\ni c$ of $[a,b]$, $V(f,P)=V(f,P\cap[a,c])+V(f,P\cap[c,b])$, and every partition may be refined to contain c. Taking suprema gives the formula, both sides being finite or infinite together. ∎

Theorem 5.16 (5.4.3, Jordan decomposition)

$f:[a,b]\to\mathbb{R}$ is of bounded variation iff $f=g-h$ with g,h increasing.

Proof. Monotone functions are of bounded variation, hence so are their differences (Proposition 5.14). Conversely, let $g(x)=V_a^x(f)$ (with $g(a)=0$) and $h=g-f$. g increases by Theorem 5.15, and for $x<y$, $h(y)-h(x)=V_x^y(f)-(f(y)-f(x))\ge 0$. ∎

Consequently a BV function has one-sided limits everywhere and at most countably many discontinuities (Theorem 3.21, Corollary 3.22). A continuous function need not be of bounded variation: $x\sin(1/x)$ on $[0,1]$.

A **curve** is a continuous $\alpha:[a,b]\to\mathbb{R}^m$; its **length** is $\Lambda(\alpha)=\sup_P\sum\|\alpha(x_i)-\alpha(x_{i-1})\|$.

Proposition 5.17 (5.4.4)

For a curve $\alpha=(\alpha_1,\dots,\alpha_m)$ the following are equivalent:
(a) $\Lambda(\alpha)<\infty$;
(b) each α_k is of bounded variation.

Proof. For each k, $|\alpha_k(x_i)-\alpha_k(x_{i-1})|\le\|\alpha(x_i)-\alpha(x_{i-1})\|\le\sum_j|\alpha_j(x_i)-\alpha_j(x_{i-1})|$. Summing, $V(\alpha_k,P)\le\sum_i\|\Delta\alpha_i\|\le\sum_j V(\alpha_j,P)$; take suprema. ∎

Theorem 5.18 (5.4.5)

Every C^1 curve $\alpha:[a,b]\to\mathbb{R}^m$ has finite length $\Lambda(\alpha)=\int_a^b\|\alpha'(t)\|\,dt$.

Proof. For vector-valued v (integrated componentwise), $\|\int v\|\le\int\|v\|$: with $u=\int v$, $\|u\|^2=\int\langle u,v\rangle\le\|u\|\int\|v\|$ by Cauchy–Schwarz. Hence $\|\alpha(x_i)-\alpha(x_{i-1})\|=\|\int_{x_{i-1}}^{x_i}\alpha'\|\le\int_{x_{i-1}}^{x_i}\|\alpha'\|$, and $\Lambda(\alpha)\le\int\|\alpha'\|$. Conversely, α' is uniformly continuous; for $\|P\|$ small, $\|\alpha'(t)-\alpha'(x_i)\|<\epsilon$ on $[x_{i-1},x_i]$, so

$$\int_{x_{i-1}}^{x_i}\|\alpha'\|\le\|\alpha'(x_i)\|\Delta x_i+\epsilon\Delta x_i=\Big\|\int_{x_{i-1}}^{x_i}(\alpha'(t)+\alpha'(x_i)-\alpha'(t))dt\Big\|+\epsilon\Delta x_i\le\|\Delta\alpha_i\|+2\epsilon\Delta x_i.$$

Summing, $\int\|\alpha'\|\le\Lambda(\alpha)+2\epsilon(b-a)$. ∎

5.5 The Riemann–Stieltjes integral

Let $\alpha:[a,b]\to\mathbb{R}$ be increasing. With $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})$ define $U(f,P,\alpha)=\sum M_i\Delta\alpha_i$, $L(f,P,\alpha)=\sum m_i\Delta\alpha_i$, and $S(f,P,\alpha)=\sum f(t_i)\Delta\alpha_i$. As in Proposition 5.1, refinement decreases U and increases L. $f\in\mathcal{R}(\alpha)$ if $\inf_P U(f,P,\alpha)=\sup_P L(f,P,\alpha)$; the common value is $\int_a^b f\,d\alpha$.

Theorem 5.19 (5.5.1)

Let α,β be increasing and f,g bounded.
(a) If $f,g\in\mathcal{R}(\alpha)$ and $c\in\mathbb{R}$, then $f+g,cf\in\mathcal{R}(\alpha)$ with $\int(f+g)\,d\alpha=\int f\,d\alpha+\int g\,d\alpha$ and $\int cf\,d\alpha=c\int f\,d\alpha$; and $f\le g$ implies $\int f\,d\alpha\le\int g\,d\alpha$.
(b) If $f\in\mathcal{R}(\alpha)$ on $[a,b]$ and $a<c<b$, then $f\in\mathcal{R}(\alpha)$ on $[a,c]$ and $[c,b]$, and $\int_a^b=\int_a^c+\int_c^b$.
(c) If $f\in\mathcal{R}(\alpha)\cap\mathcal{R}(\beta)$ and $c\ge0$, then $f\in\mathcal{R}(\alpha+\beta)\cap\mathcal{R}(c\alpha)$ with $\int f\,d(\alpha+\beta)=\int f\,d\alpha+\int f\,d\beta$ and $\int f\,d(c\alpha)=c\int f\,d\alpha$.
(d) If $f\in\mathcal{R}(\alpha)$, then $|f|\in\mathcal{R}(\alpha)$ and $|\int f\,d\alpha|\le\int|f|\,d\alpha\le\sup|f|\,(\alpha(b)-\alpha(a))$.

Proof. The proofs of Theorems 5.3, 5.4 and 5.5 apply verbatim with Δx_i replaced by $\Delta\alpha_i\ge0$. For (c), $U(f,P,\alpha+\beta)=U(f,P,\alpha)+U(f,P,\beta)$ and likewise for L. ∎

Theorem 5.20 (5.5.2)

For bounded f and increasing α the following are equivalent:
(a) $f\in\mathcal{R}(\alpha)$;
(b) for every $\epsilon>0$ there is P with $U(f,P,\alpha)-L(f,P,\alpha)<\epsilon$;
(c) there is A such that for every $\epsilon>0$ some P_0 has $|S(f,P,\alpha)-A|<\epsilon$ for every $P\supset P_0$ and every choice of tags. In that case $A=\int f\,d\alpha$.

Proof. (a)⇔(b) as in Proposition 5.2.

(b)⇒(c): with $A=\int f\,d\alpha$ and P_0 as in (b), for $P\supset P_0$ both $S(f,P,\alpha)$ and A lie in $[L(f,P,\alpha),U(f,P,\alpha)]$, an interval of length $<\epsilon$.

(c)⇒(b): for $P\supset P_0$, taking suprema and infima over tags, $U(f,P,\alpha)\le A+\epsilon$ and $L(f,P,\alpha)\ge A-\epsilon$. ∎

Example 5.3

(a) If α is the unit step at $c\in(a,b)$ ($\alpha=0$ on $[a,c]$, 1 on $(c,b]$) and f is continuous at c, then $\int f\,d\alpha=f(c)$. If f and α are both discontinuous from the right at c, f may fail to be integrable.

(b) For a step function α with jumps c_k at points s_k and continuous f, $\int f\,d\alpha = \sum c_k f(s_k)$: sums are a special case of Stieltjes integrals.

For α of bounded variation, write $\alpha = \alpha_1 - \alpha_2$ with α_i increasing (Theorem 5.16). If $f \in \mathcal{R}(\alpha_1) \cap \mathcal{R}(\alpha_2)$ for some such decomposition, we say $f \in \mathcal{R}(\alpha)$ and set $\int f\,d\alpha = \int f\,d\alpha_1 - \int f\,d\alpha_2$; by Theorem 5.22 below this does not depend on the decomposition. The Riemann–Stieltjes sum $S(f, P, \alpha)$ is defined as before.

Proposition 5.21 (5.5.3)

If f is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ for every α of bounded variation.

Proof. It suffices to treat increasing α. Given ϵ, by uniform continuity choose P with $M_i - m_i < \epsilon$ for all i; then $U(f, P, \alpha) - L(f, P, \alpha) \le \epsilon(\alpha(b) - \alpha(a))$. ∎

Theorem 5.22 (5.5.4)

For bounded f and α of bounded variation the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = A$;
(b) for every $\epsilon > 0$ there is P_0 with $|S(f, P, \alpha) - A| < \epsilon$ for all $P \supset P_0$ and all tags.

Proof. (a)⇒(b): $S(f, P, \alpha) = S(f, P, \alpha_1) - S(f, P, \alpha_2)$; apply Theorem 5.20(c) to each and take a common refinement. In particular A is determined by f and α, so the integral is well defined.

(b)⇒(a): let $V(x) = V_a^x(\alpha)$; we show $f \in \mathcal{R}(V)$. Let $K = \sup|f|$ and $\epsilon > 0$. Choose P_0 as in (b) and refine it so that $V(b) - \sum_i |\Delta\alpha_i| < \epsilon$ (definition of $V(b)$; refining keeps this). For $P \supset P_0$ write

$$U(f, P, V) - L(f, P, V) = \sum (M_i - m_i)(\Delta V_i - |\Delta\alpha_i|) + \sum (M_i - m_i)|\Delta\alpha_i|.$$

Since $\Delta V_i \ge |\Delta\alpha_i|$, the first sum is at most $2K\epsilon$. For the second, choose s_i, t_i in the i-th interval with $f(t_i) - f(s_i) > M_i - m_i - \epsilon$, and swap them where $\Delta\alpha_i < 0$; then the second sum is less than $\sum (f(t_i) - f(s_i))\Delta\alpha_i + \epsilon V(b) = S(f, P, \alpha) - S'(f, P, \alpha) + \epsilon V(b) < 2\epsilon + \epsilon V(b)$, where S, S' are sums with tags t_i, s_i. By Theorem 5.20, $f \in \mathcal{R}(V)$. Then the sums $S(f, P, V - \alpha) = S(f, P, V) - S(f, P, \alpha)$ satisfy the condition of Theorem 5.20(c) for the increasing function $V - \alpha$, so $f \in \mathcal{R}(V - \alpha)$, and with the decomposition $\alpha = V - (V - \alpha)$, $\int f\,d\alpha = A$. ∎

Theorem 5.23 (5.5.5)

Let α be differentiable with α' continuous on $[a, b]$. If f is bounded and $f \in \mathcal{R}(\alpha)$, then $f\alpha'$ is Riemann integrable and $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. α is of bounded variation (Lipschitz). By the mean value theorem $\Delta\alpha_i = \alpha'(u_i)\Delta x_i$ with $u_i \in (x_{i-1}, x_i)$, so

$$\left| S(f, P, \alpha) - \sum f(t_i)\alpha'(t_i)\Delta x_i \right| \le K \sum |\alpha'(u_i) - \alpha'(t_i)|\Delta x_i.$$

By uniform continuity of α' this is $\le K\epsilon(b - a)$ once $\|P\|$ is small. Combining with Theorem 5.22(b), the Riemann sums of $f\alpha'$ over all sufficiently fine refinements of some P_0 are within $\epsilon + K\epsilon(b - a)$ of $\int f\,d\alpha$. By Theorem 5.20 with $\alpha(x) = x$, $f\alpha' \in \mathcal{R}$ with the stated integral. ∎

Theorem 5.24 (5.5.6, integration by parts)

Let f, α be of bounded variation on $[a, b]$. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ and $\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a)$.

Proof. Let $A = \int f\,d\alpha$ and P_0 as in Theorem 5.22(b). Let $Q \supset P_0$ be a partition $a = y_0 < \cdots < y_m = b$ with tags $s_k \in [y_{k-1}, y_k]$. By summation by parts,

$$S(\alpha, Q, f) = \sum_k \alpha(s_k)(f(y_k) - f(y_{k-1})) = f(b)\alpha(b) - f(a)\alpha(a) - \sum_{k=0}^{m} f(y_k)(\alpha(s_{k+1}) - \alpha(s_k)),$$

with $s_0 = a$, $s_{m+1} = b$. The last sum is a Riemann–Stieltjes sum $S(f, P, \alpha)$ for the partition P formed by the points s_k together with the y_k (zero-length intervals contribute nothing), with tags y_k; $P \supset Q \supset P_0$. So $|S(\alpha, Q, f) - (f(b)\alpha(b) - f(a)\alpha(a) - A)| < \epsilon$, and Theorem 5.22 applies to α and f. ■

Corollary 5.25 (5.5.7, second mean value theorem)

If $f : [a, b] \to \mathbb{R}$ is increasing and g is continuous, there is $c \in [a, b]$ with $\int_a^b fg = f(a)\int_a^c g + f(b)\int_c^b g$.

Proof. Let $G(x) = \int_a^x g$, a C^1 function with $G' = g$. $G \in \mathcal{R}(f)$ (Proposition 5.21), so $f \in \mathcal{R}(G)$ and $\int f\,dG = f(b)G(b) - \int G\,df$ (Theorem 5.24), while $\int f\,dG = \int fg$ (Theorem 5.23). With m, M the minimum and maximum of G, $m(f(b) - f(a)) \le \int G\,df \le M(f(b) - f(a))$, so $\int G\,df = G(c)(f(b) - f(a))$ for some c by the intermediate value theorem. Hence $\int fg = f(b)G(b) - G(c)(f(b) - f(a)) = f(a)G(c) + f(b)(G(b) - G(c))$. ■

Appendix A. The Number Systems

Chapter 1 took the real numbers as a complete ordered field. Here we construct such a field. We accept the arithmetic and order of $\mathbb{N} = \{0, 1, 2, \dots\}$, including the fact that every nonempty subset of $\mathbb{N}$ has a least element, build $\mathbb{Z}$ and $\mathbb{Q}$ by equivalence relations, and obtain $\mathbb{R}$ from Cauchy sequences of rationals.

A.1 Integers and rationals

On $\mathbb{N} \times \mathbb{N}$ put $(m, n) \sim (m', n') \iff m + n' = n + m'$ (think of $m - n$); this is an equivalence relation, and $(m, n) \sim (m - k, n - k)$ when $m, n \geq k$. $\mathbb{Z}$ is the set of classes $[m, n]$, with $[m, n] \geq [k, l] \iff m + l \geq n + k$, $[m, n] + [k, l] = [m + k, n + l]$, $[m, n] \cdot [k, l] = [mk + nl, ml + nk]$. These are well defined (independent of representatives), and every integer is $[n, 0]$, $[0, 0]$ or $[0, n]$ for a unique $n \geq 1$, written n, 0, $-n$; the map $n \mapsto [n, 0]$ embeds $\mathbb{N}$ preserving operations and order.

Theorem A.1 (Theorem 1)

(a) For $a, b \in \mathbb{Z}$, $a \geq b$ or $b \geq a$.
(b) A nonempty subset of $\mathbb{Z}$ bounded above has a largest element, and one bounded below has a least element.

Proof. (a) reduces to the same property of $\mathbb{N}$ for $m + l$ and $n + k$.

(b) If S is bounded below by b, then $\{s - b : s \in S\} \subset \mathbb{N}$ has a least element k, and $b + k$ is least in S; for S bounded above apply this to $-S$. ∎

$\mathbb{Z}$ satisfies the field axioms except the existence of multiplicative inverses, and with $P_{\mathbb{Z}} = \{1, 2, \dots\}$ the order axioms. On $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ put $(a, b) \sim (c, d) \iff ad = bc$ and order the classes by $(a, b) > (c, d) \iff abd^2 > cdb^2$; the classes are the rationals a/b, with $a/b + c/d = (ad + bc)/bd$ and $(a/b)(c/d) = ac/bd$. One checks that $\mathbb{Q}$ is an ordered field containing $\mathbb{Z}$, so the results of Section 1.1, valid in any ordered field, hold in $\mathbb{Q}$.

A.2 Real numbers as classes of Cauchy sequences

A rational sequence $(a(i))$ is a **fundamental sequence** if for every rational $\epsilon > 0$ there is N with $|a(i) - a(j)| < \epsilon$ for $i, j \geq N$.

Theorem A.2 (Theorem 2)

(a) Fundamental sequences are bounded.
(b) A bounded monotone rational sequence is fundamental.
(c) If $(a(i))$ is fundamental, so is $(|a(i)|)$.
(d) If $(a(i)), (b(i))$ are fundamental, so are $(a(i) \pm b(i))$ and $(a(i)b(i))$.
(e) If $(a(i))$ is fundamental and there are N and a rational $d > 0$ with $|a(i)| \geq d$ for $i \geq N$, then $(1/a(i))_{i \geq N}$ is fundamental.

Proof. (a) as in Proposition 2.12.

(b) If an increasing sequence bounded by B were not fundamental, there would be a rational $\epsilon > 0$ and indices $i_1 < j_1 \leq i_2 < j_2 \leq \cdots$ with $a(j_k) - a(i_k) \geq \epsilon$; then $a(j_k) \geq a(i_1) + k\epsilon > B$ for large k.

(c) $\big||a(i)| - |a(j)|\big| \leq |a(i) - a(j)|$.

(d) $|a(i)b(i) - a(j)b(j)| \leq |a(i)||b(i) - b(j)| + |b(j)||a(i) - a(j)|$ and (a).

(e) $|1/a(i) - 1/a(j)| \le |a(i) - a(j)|/d^2$. ■

Two fundamental sequences are **equivalent**, $(a(i)) \sim (b(i))$, if for every rational $\epsilon > 0$ there is N with $|a(i) - b(i)| < \epsilon$ for $i \ge N$; this is an equivalence relation. A **real number** is an equivalence class $[a]$; $\mathbb{R}$ is the set of classes. Define $[a] > [b]$ if there are a rational $d > 0$ and N with $a(i) - b(i) > d$ for $i \ge N$; this does not depend on the representatives.

Theorem A.3 (Theorem 3, trichotomy)

For $[a], [b] \in \mathbb{R}$ exactly one of $[a] > [b]$, $[a] = [b]$, $[b] > [a]$ holds.

Proof. For rational $\epsilon > 0$ let N_ϵ be such that $|a(i) - a(N_\epsilon)| < \epsilon$ and $|b(i) - b(N_\epsilon)| < \epsilon$ for $i \ge N_\epsilon$. Then for $i \ge N_\epsilon$,

$$d_\epsilon := a(N_\epsilon) - b(N_\epsilon) - 2\epsilon < a(i) - b(i) < a(N_\epsilon) - b(N_\epsilon) + 2\epsilon =: d'_\epsilon.$$

If $d_\epsilon > 0$ for some ϵ, then $[a] > [b]$; if $d'_\epsilon < 0$ for some ϵ, then $[b] > [a]$. Otherwise for every ϵ, $|a(N_\epsilon) - b(N_\epsilon)| \le 2\epsilon$, so $|a(i) - b(i)| < 4\epsilon$ for $i \ge N_\epsilon$, and $[a] = [b]$. Two of the cases cannot hold together, by the definitions. ■

Define $[a]+[b] = [a+b]$ and $[a][b] = [ab]$ (termwise operations); these are well defined by Theorem A.2. For $r \in \mathbb{Q}$ let r^* be the class of the constant sequence r; $r \mapsto r^*$ embeds $\mathbb{Q}$ in $\mathbb{R}$ preserving operations and order. 0^* and 1^* are the identities, $[-a]$ is the additive inverse, and if $[a] \ne 0^*$ then by Theorem A.3 there are $d > 0$ and N with $|a(i)| \ge d$ for $i \ge N$, so the class of $(1/a(i))_{i \ge N}$ (Theorem A.2(e)) is the inverse. With $P = \{[a] : [a] > 0^*\}$, $\mathbb{R}$ is an ordered field, and $|[a]| = [|a|]$.

Theorem A.4 (Theorem 4, density of $\mathbb{Q}$)

If $[b] > [a]$, there is a rational r with $[b] > r^* > [a]$.

Proof. Choose N_1 and rational $d > 0$ with $b(i) - a(i) > d$ for $i \ge N_1$, and N_2 with $|a(i) - a(j)| < d/4$, $|b(i) - b(j)| < d/4$ for $i, j \ge N_2$; let $N = \max(N_1, N_2)$ and $r = a(N) + d/2$. For $i \ge N$, $r - a(i) = d/2 + (a(N) - a(i)) > d/4$, so $r^* > [a]$; and $b(i) - r = (b(i) - b(N)) + (b(N) - a(N)) - d/2 > -d/4 + d - d/2 = d/4$, so $[b] > r^*$. ■

Theorem A.5 (Theorem 5, completeness)

Every Cauchy sequence $([a_n])$ in $\mathbb{R}$ converges in $\mathbb{R}$.

Proof. If the sequence is eventually constant there is nothing to prove; otherwise, passing to a subsequence (which suffices by Lemma 2.13, whose proof is valid in any ordered field), assume $[a_n] \ne [a_{n+1}]$ for all n. By Theorem A.4 choose rationals $r(n)$ strictly between $[a_n]$ and $[a_{n+1}]$; then $|[a_n] - r(n)^*| < |[a_n] - [a_{n+1}]| \to 0$. The rational sequence $(r(n))$ is fundamental, since $|r(n) - r(m)|^* \le |r(n)^* - [a_n]| + |[a_n] - [a_m]| + |[a_m] - r(m)^*|$. Let $[r]$ be its class. Given $[e] > 0$ choose, by Theorem A.4, a rational $\epsilon > 0$ with $\epsilon^* < [e]$, and M with $|r(i) - r(j)| < \epsilon/2$ for $i, j \ge M$. For $n \ge M$, $|r(n)^* - [r]| \le (\epsilon/2)^*$, because the termwise differences $r(n) - r(i)$ are eventually below $\epsilon/2$. Hence for large n, $|[a_n] - [r]| \le |[a_n] - r(n)^*| + (\epsilon/2)^* < [e]$. ■

Finally, Theorems A.4 and A.5 imply the completeness axiom: given S nonempty and bounded above, the bisection argument after Theorem 1.20 produces nested intervals with rational end points whose lengths tend to 0; the left end points form a Cauchy sequence, whose limit (Theorem A.5) is $\sup S$. So $\mathbb{R}$ is a complete ordered field, unique up to isomorphism by Theorem 1.10.